

Contact process

This is the industrial process for making sulfuric acid. Sulfur dioxide gas reacts with water using a catalyst to produce the acid in a multistage process. Sulfuric acid is one of the most powerful acids. It is used in car batteries and its salts (the sulfates) are used in fertilizers. It is also used in papermaking.

1. Furnace

In the furnace, sulfur (S) is burned with oxygen (O) from the air to form sulfur dioxide (SO₂).

2. Cleaning the gas

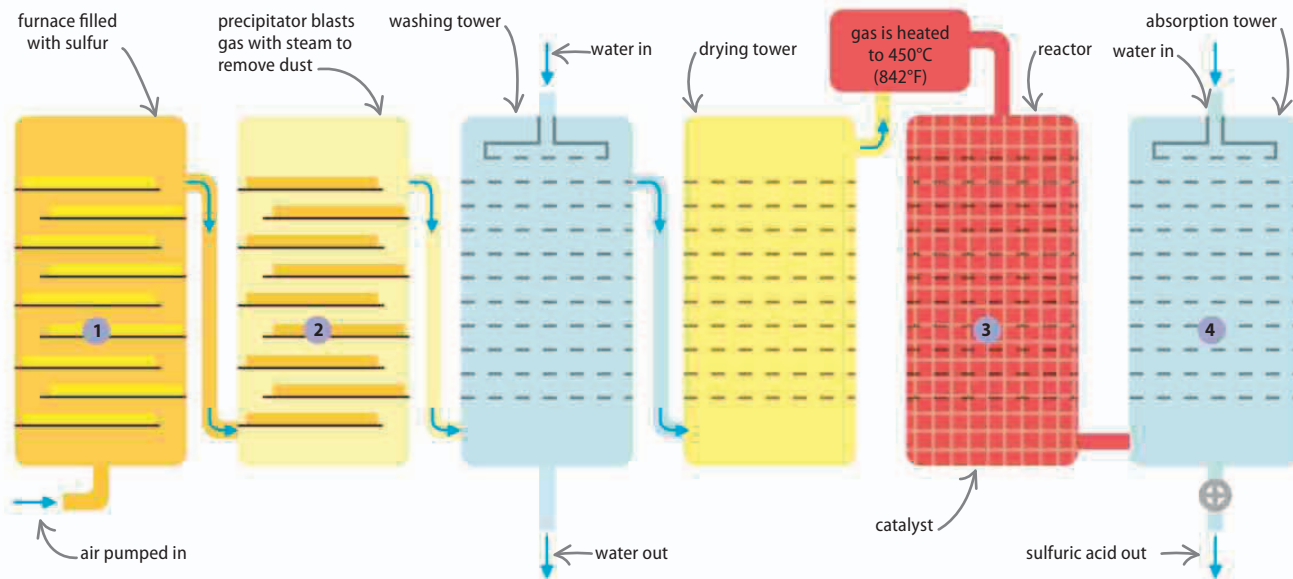
In the next three chambers, the gas is filtered, washed, and dried to remove any impurities that could interfere with the catalyst.

3. Reactor

The vanadium oxide catalyst makes the sulfur dioxide react with more oxygen to form sulfur trioxide (SO₃).

4. Absorption tower

The sulfur trioxide is dissolved in a little sulfuric acid. This makes it safe to dilute with water to make a lot more sulfuric acid—the end product.



Downs cells

Pure chlorine gas (Cl₂) and sodium metal (Na) are made by the electrolysis of sodium chloride (common salt: NaCl). This takes place on an industrial scale in a large tank called a Downs cell. The tank contains liquid salt—it is heated to more than 600°C (1,112°F) so that the salt melts. When an electric current is run through the liquid, the molten salt breaks up into sodium and chloride ions, which move to the electrodes and turn into atoms. The pure elements can then be collected.

1. At the iron cathode ($2\text{Na}^+ + 2\text{e}^- \rightarrow 2\text{Na}$)

Positive sodium ions (Na⁺) move to the cathode, where they gain an electron each to form sodium atoms (Na). This metal is less dense than the sodium chloride electrolyte so it floats to the surface where it can be collected.

2. At the carbon anode ($2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$)

Negative chlorine ions (Cl⁻) are attracted to the positively charged anode, where they lose an electron each to form chlorine atoms (Cl). The element bubbles out of the liquid electrolyte as chlorine gas (Cl₂).

